Enzyme immobilisation on MANAE-agarose: support preparation, strategies A and B, and yield evaluation

What you're going to get / What you expect

An immobilised enzyme derivative on a MANAE-agarose support, prepared by the two strategies compared here, with the immobilisation yield quantified for each. Ionic adsorption followed by crosslinking (Strategy B) keeps about ten times more active enzyme than direct covalent attachment (Strategy A). Glutaraldehyde on this model support plays the same amino-crosslinking role that genipin plays on the final alginate-chitosan sphere.

Materials list (see each section)

- Support preparation (per 20 g BCL agarose): BCL agarose bead standard; glycidol; sodium periodate; ethylenediamine; sodium borohydride; NaOH; carbonate buffer (>100 mM, pH 9); acetate buffer (>100 mM, pH 4); distilled water.
- Immobilisation: MANAE-agarose support; glutaraldehyde; phosphate buffer 200 mM pH 7; potassium phosphate buffer 25 mM pH 7; sodium phosphate buffer 25 mM pH 7; Lecitase Ultra solution; Triton X-100 (0.1% v/v, detergent variable).
- Yield evaluation: microcentrifuge (604 x g); pNPB continuous assay set-up (348 nm); a free-enzyme aliquot for the inactivation control.

Steps

SUPPORT PREPARATION

1. Dissolve 20 g of filtered, washed BCL agarose in 4 mL of distilled water.
2. Add a reducing alkaline solution of 0.45 g NaOH plus 0.19 g sodium borohydride in 6.7 mL of water.
3. In an ice bath, slowly drip 5 mL of glycidol to introduce epoxide groups; keep under agitation overnight.
4. Vacuum filter and wash; oxidise to glyoxyl-agarose with 10 mL of aqueous solution containing 90 umol of sodium periodate per gram of dry support; agitate 3 hours and wash thoroughly.
5. Aminate by adding 4 mL of 2 M ethylenediamine at pH 10 per gram of glyoxyl-agarose, in an ice bath to prevent exothermic overheating.
6. Reduce the Schiff bases with sodium borohydride to 10 mg/mL under agitation for 2 hours.
7. Wash sequentially with 500 mL carbonate buffer (>100 mM, pH 9), 500 mL acetate buffer (>100 mM, pH 4) and plenty of distilled water; dry, weigh and store at 4 C.

STRATEGY A (covalent on pre-activated support)

1. Suspend 1 g of MANAE-agarose in 4 mL of 200 mM phosphate buffer pH 7 and add 10 mL of 10% glutaraldehyde; keep under agitation overnight to activate the support.
2. Wash the activated support with distilled water and use it immediately (standing oxidises the reactive aldehyde groups); incubate with Lecitase Ultra at 1 mg enzyme per gram of support. Run in parallel in standard buffer and with 0.1% (v/v) Triton X-100.

STRATEGY B (ionic adsorption and subsequent crosslinking)

1. Suspend 10 g of unactivated MANAE-agarose in 20 mL of 25 mM potassium phosphate buffer pH 7; add Lecitase Ultra at 1 mg enzyme per gram of support and keep under gentle stirring at 25 ºC, following adsorption by the fall in supernatant activity.
2. Once adsorbed, crosslink in 25 mM sodium phosphate buffer pH 7 with 0.5% (v/v) glutaraldehyde under gentle stirring for 1 hour at 25 ºC; filter and wash to remove free glutaraldehyde and enzyme in suspension; mature for a further 20 hours at 25 ºC to consolidate a multipoint covalent bond. Run in parallel with and without 0.1% (v/v) Triton X-100.

YIELD EVALUATION

3. At each time point, measure the activity of a direct aliquot of the homogeneous suspension (free plus bound enzyme), Act(ET), by the pNPB assay at 348 nm.
4. Take a second aliquot, centrifuge at 604 x g for 2 minutes to sediment the agarose, and measure the clarified supernatant (free enzyme only), Act(ES).
5. Compute the immobilised activity and the yield with the equations below; run a free-enzyme control (no support) in parallel to confirm that activity loss is due to immobilisation and not to buffer effects or thermal denaturation.

Immobilised activity and yield:

$$Act(Ei) = Act(ET) - Act(ES)$$

$$Yield\ (\%) = [\ Act(Ei) / Act(ET)\ ] \times 100$$

Troubleshooting

- Strategy A, use the activated support immediately: standing oxidises the aldehyde groups and lowers coupling. Low activity without detergent here reflects the closed enzyme being anchored through the lid-adjacent lysines (Lys24, Lys259), jamming the lid shut; 0.1% Triton X-100 holds the lid open and raises retention.
- Strategy B works best without detergent: the enzyme docks through its acidic crown (Asp27, Glu56, Asp57, Asp62) with the active site projected outward; Triton X-100 scrambles that orientation and the crosslinking then freezes a low-activity arrangement. Do not skip the 20 hour maturation.
- Similar yields can hide very different retained activities: report retained activity, not just percentage bound, and never omit the free-enzyme control.